

Eshwar Pamula

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EDUCATION

The Ohio State University

Bachelor of Science in Mechanical Engineering with Honors Research Distinction

Minor: Economics

Columbus, OH

Graduation: December 2025

GPA: 3.36/4.0

PROFESSIONAL EXPERIENCE

Flux Marine

Powertrain Engineering Intern

Bristol, RI

May 2025 - August 2025

- Designed and built a specialized shrink-fit jig for stator core assembly, reducing process time while ensuring all segments are seated flush in the motor housing.
- Contributed to product development from prototype through production readiness across three different motor builds at a Series-A marine propulsion startup.
- Conducted dynamometer testing across torque and RPM ranges to validate motor performance and inform design iterations.
- Optimized assembly process for segmented stator cores, ensuring consistent tolerances across all production-ready parts.

Allegion Inc.

R&D Mechanical Engineering Intern

Colorado Springs, CO

May 2024 - August 2024

- Worked on new product development initiatives for two new electro-mechanical residential lock set designs.
- Engineered a new CAM assembly using PTC Creo for commercial Mortise locks, improving operational efficiency.
- Designed an interconnect assembly featuring two different CAMs and tested using ANSYS to adhere to BHMA standards.
- Manufactured early prototypes (3D printing and CNC) of the lock system, ensuring feasibility for the 2025 product pipeline.

Ecosystem Energy

Mechanical Engineering Intern

Columbus, OH

May 2023 - August 2023

- Monitored and analyzed daily control data from heat exchangers to enhance energy efficiency and system optimization.
- Led a \$300K project to install occupancy sensors in a campus facility, integrating dynamic energy control via pneumatic valves and achieving a 27% reduction in energy consumption.

Engie North America

Project Engineering Intern

Columbus, OH

October 2022 - May 2023

- Conducted comprehensive engineering reviews of HVAC designs for Ohio State's Combined Heat and Power plant.

RESEARCH & PROJECT EXPERIENCE

Multi-functional Materials and Intelligent Design Lab

Undergraduate Researcher

Columbus, OH

May 2024 - Present

- Awarded College of Engineering (Honors Distinction) grant for AI-driven geometric primitive-based topology optimization, creating a novel approach that extracts meaningful structural patterns from optimized designs.

Gear and Power Transmission Laboratory

Undergraduate Research Assistant

Columbus, OH

April 2023 - May 2024

- Conducted experimental analysis of polymer (Torlon) gear properties using Symbrium 3-shaft-bending machines.
- Monitored and controlled oil flow, temperature, and torque conditions to assess thermal performance of gear systems.

LEADERSHIP AND INVOLVEMENT

Formula Buckeyes FSAE

Suspension Engineer

Columbus, OH

2022 - August 2024

- Designed and manufactured uprights, mounts, and hard points using the mill and lathe for the control arms and sway bar.
- Helped manufacture in-house Billet 7075-T6 Uprights using Kia SKT25 CNC to reduce stiffness and complexity.
- Contributed to designing and manufacturing a rear Anti-Roll Bar (T-Bar setup) with SolidWorks and Huron Lathe CNC.
- Played a key role in the team's success, securing a 3rd-place finish in 2023 and a 1st-place finish in 2024 at Michigan FSAE.

Ohio State University: College of Mechanical and Aerospace Engineering

Undergraduate Teaching Assistant

Columbus, OH

April 2023 - Present

- Taught and mentored 400+ students over five semesters in Statics (ME2010) and Mechanics of Materials (ME2020).
- Runner-Up for the Best MAE Undergraduate Teaching Assistant in 2025.

AWARDS, SKILLS, AND INTERESTS

- Awards:** ShowOHIO 3rd place ('23), Ohio State Fintech Pitch Competition Winner ('23), OSHA 10 Certification
- CAD & FEA:** SolidWorks, CATIA, CREO, AutoCAD, ANSYS
- Skills & Tools:** CNC Machining, Finite Element Analysis (FEA), 3D Printing, Lathe, Mill
- Programming Languages:** MATLAB, Python, Arduino IDE